

Additional Topics

2.A Homology and Fundamental Group

There is a close connection between $H_1(X)$ and $\pi_1(X)$, arising from the fact that a map $f: I \rightarrow X$ can be viewed as either a path or a singular 1-simplex. If f is a loop, with $f(0) = f(1)$, this singular 1-simplex is a cycle since $\partial f = f(1) - f(0)$.

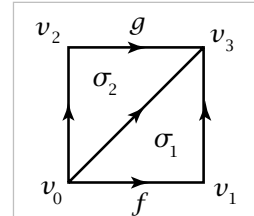
Theorem 2A.1. *By regarding loops as singular 1-cycles, we obtain a homomorphism $h: \pi_1(X, x_0) \rightarrow H_1(X)$. If X is path-connected, then h is surjective and has kernel the commutator subgroup of $\pi_1(X)$, so h induces an isomorphism from the abelianization of $\pi_1(X)$ onto $H_1(X)$.*

Proof: Recall the notation $f \simeq g$ for the relation of homotopy, fixing endpoints, between paths f and g . Regarding f and g as chains, the notation $f \sim g$ will mean that f is homologous to g , that is, $f - g$ is the boundary of some 2-chain. Here are some facts about this relation.

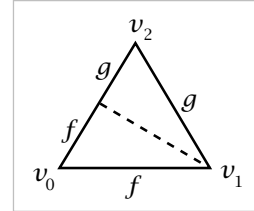
(i) If f is a constant path, then $f \sim 0$. Namely, f is a cycle since it is a loop, and since $H_1(\text{point}) = 0$, f must then be a boundary. Explicitly, f is the boundary of the constant singular 2-simplex σ having the same image as f since

$$\partial \sigma = \sigma|_{[v_1, v_2]} - \sigma|_{[v_0, v_2]} + \sigma|_{[v_0, v_1]} = f - f + f = f$$

(ii) If $f \simeq g$ then $f \sim g$. To see this, consider a homotopy $F: I \times I \rightarrow X$ from f to g . This yields a pair of singular 2-simplices σ_1 and σ_2 in X by subdividing the square $I \times I$ into two triangles $[v_0, v_1, v_3]$ and $[v_0, v_2, v_3]$ as shown in the figure. When one computes $\partial(\sigma_1 - \sigma_2)$, the two restrictions of F to the diagonal of the square cancel, leaving $f - g$ together with two constant singular 1-simplices from the left and right edges of the square. By (i) these are boundaries, so $f - g$ is also a boundary.



(iii) $f \cdot g \sim f + g$, where $f \cdot g$ denotes the product of the paths f and g . For if $\sigma: \Delta^2 \rightarrow X$ is the composition of orthogonal projection of $\Delta^2 = [v_0, v_1, v_2]$ onto the edge $[v_0, v_2]$ followed by $f \cdot g: [v_0, v_2] \rightarrow X$, then $\partial \sigma = g - f \cdot g + f$.



(iv) $\bar{f} \sim -f$, where $\bar{f}$ is the inverse path of f . This follows from the preceding three observations, which give $f + \bar{f} \sim f \cdot \bar{f} \sim 0$.

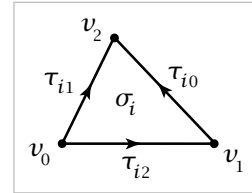
Applying (ii) and (iii) to loops, it follows that we have a well-defined homomorphism $h: \pi_1(X, x_0) \rightarrow H_1(X)$ sending the homotopy class of a loop f to the homology class of the 1-cycle f .

To show h is surjective when X is path-connected, let $\sum_i n_i \sigma_i$ be a 1-cycle representing a given element of $H_1(X)$. After relabeling the σ_i 's we may assume each n_i is ± 1 . By (iv) we may in fact take each n_i to be $+1$, so our 1-cycle is $\sum_i \sigma_i$. If some σ_i is not a loop, then the fact that $\partial(\sum_i \sigma_i) = 0$ means there must be another σ_j such that the composed path $\sigma_i \cdot \sigma_j$ is defined. By (iii) we may then combine the terms σ_i and σ_j into a single term $\sigma_i \cdot \sigma_j$. Iterating this, we reduce to the case that each σ_i is a loop. Since X is path-connected, we may choose a path γ_i from x_0 to the basepoint of σ_i . We have $\gamma_i \cdot \sigma_i \cdot \bar{\gamma}_i \sim \sigma_i$ by (iii) and (iv), so we may assume all σ_i 's are loops at x_0 . Then we can combine all the σ_i 's into a single σ by (iii). This says the given element of $H_1(X)$ is in the image of h .

The commutator subgroup of $\pi_1(X)$ is contained in the kernel of h since $H_1(X)$ is abelian. To obtain the reverse inclusion we will show that every class $[f]$ in the kernel of h is trivial in the abelianization $\pi_1(X)_{ab}$ of $\pi_1(X)$.

If an element $[f] \in \pi_1(X)$ is in the kernel of h , then f , as a 1-cycle, is the boundary of a 2-chain $\sum_i n_i \sigma_i$. Again we may assume each n_i is ± 1 . As in the discussion preceding Proposition 2.6, we can associate to the chain $\sum_i n_i \sigma_i$ a 2-dimensional Δ -complex K by taking a 2-simplex Δ_i^2 for each σ_i and identifying certain pairs of edges of these 2-simplices. Namely, if we apply the usual boundary formula to write $\partial \sigma_i = \tau_{i0} - \tau_{i1} + \tau_{i2}$ for singular 1-simplices τ_{ij} , then the formula

$$f = \partial(\sum_i n_i \sigma_i) = \sum_i n_i \partial \sigma_i = \sum_{i,j} (-1)^j n_i \tau_{ij}$$

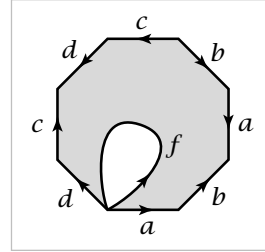


implies that we can group all but one of the τ_{ij} 's into pairs for which the two coefficients $(-1)^j n_i$ in each pair are $+1$ and -1 . The one remaining τ_{ij} is equal to f . We then identify edges of the Δ_j^2 's corresponding to the paired τ_{ij} 's, preserving orientations of these edges so that we obtain a Δ -complex K .

The maps σ_i fit together to give a map $\sigma: K \rightarrow X$. We can deform σ , staying fixed on the edge corresponding to f , so that each vertex maps to the basepoint x_0 , in the following way. Paths from the images of these vertices to x_0 define such a homotopy on the union of the 0-skeleton of K with the edge corresponding to f , and then we can appeal to the homotopy extension property in Proposition 0.16 to extend this homotopy to all of K . Alternatively, it is not hard to construct such an extension by hand. Restricting the new σ to the simplices Δ_i^2 , we obtain a new chain $\sum_i n_i \sigma_i$ with boundary equal to f and with all τ_{ij} 's loops at x_0 .

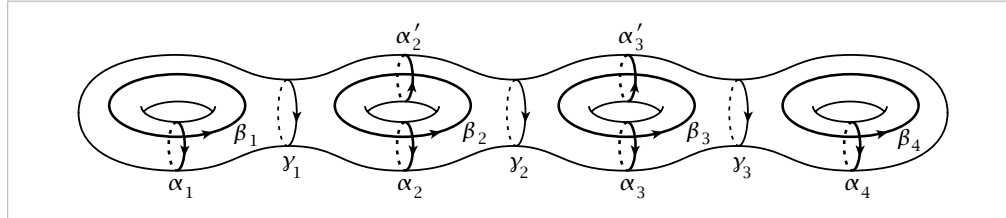
Using additive notation in the abelian group $\pi_1(X)_{ab}$, we have the formula $[f] = \sum_{i,j} (-1)^j n_i [\tau_{ij}]$ because of the canceling pairs of τ_{ij} 's. We can rewrite the summation $\sum_{i,j} (-1)^j n_i [\tau_{ij}]$ as $\sum_i n_i [\partial \sigma_i]$ where $[\partial \sigma_i] = [\tau_{i0}] - [\tau_{i1}] + [\tau_{i2}]$. Since σ_i gives a nullhomotopy of the composed loop $\tau_{i0} - \tau_{i1} + \tau_{i2}$, we conclude that $[f] = 0$ in $\pi_1(X)_{ab}$. $\square$

The end of this proof can be illuminated by looking more closely at the geometry. The complex K is in fact a compact surface with boundary consisting of a single circle formed by the edge corresponding to f . This is because any pattern of identifications of pairs of edges of a finite collection of disjoint 2-simplices produces a compact surface with boundary. We leave it as an exercise for the reader to check that the algebraic formula $f = \partial(\sum_i n_i \sigma_i)$ with each $n_i = \pm 1$ implies that K is an orientable surface. The component of K containing the boundary circle is a standard closed orientable surface of some genus g with an open disk removed, by the basic structure theorem for compact orientable surfaces. Giving this surface the cell structure indicated in the figure, it then becomes obvious that f is homotopic to a product of g commutators in $\pi_1(X)$.

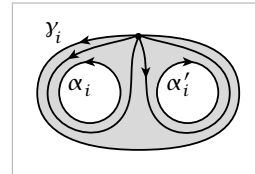


The map $h: \pi_1(X, x_0) \rightarrow H_1(X)$ can also be defined by $h([f]) = f_*(\alpha)$ where $f: S^1 \rightarrow X$ represents a given element of $\pi_1(X, x_0)$, f_* is the induced map on H_1 , and α is the generator of $H_1(S^1) \approx \mathbb{Z}$ represented by the standard map $\sigma: I \rightarrow S^1$, $\sigma(s) = e^{2\pi is}$. This is because both $[f] \in \pi_1(X, x_0)$ and $f_*(\alpha) \in H_1(X)$ are represented by the loop $f\sigma: I \rightarrow X$. A consequence of this definition is that $h([f]) = h([g])$ if f and g are homotopic maps $S^1 \rightarrow X$, since $f_* = g_*$ by Theorem 2.10.

Example 2A.2. For the closed orientable surface M of genus g , the abelianization of $\pi_1(M)$ is $\mathbb{Z}^{2g}$, the product of $2g$ copies of $\mathbb{Z}$, and a basis for $H_1(M)$ consists of the 1-cycles represented by the 1-cells of M in its standard CW structure. We can also represent a basis by the loops α_i and β_i shown in the figure below since these



loops are homotopic to the loops represented by the 1-cells, as one can see in the picture of the cell structure in Chapter 0. The loops γ_i , on the other hand, are trivial in homology since the portion of M on one side of γ_i is a compact surface bounded by γ_i , so γ_i is homotopic to a loop that is a product of commutators, as we saw a couple paragraphs earlier. The loop α'_i represents the same homology class as α_i since the region between γ_i and $\alpha_i \cup \alpha'_i$ provides a homotopy between γ_i and a product of two loops homotopic to α_i and the inverse of α'_i , so $\alpha_i - \alpha'_i \sim \gamma_i \sim 0$, hence $\alpha_i \sim \alpha'_i$.



2.B Classical Applications

In this section we use homology theory to prove several interesting results in topology and algebra whose statements give no hint that algebraic topology might be involved.

To begin, we calculate the homology of complements of embedded spheres and disks in a sphere. Recall that an embedding is a map that is a homeomorphism onto its image.

Proposition 2B.1. (a) *For an embedding $h: D^k \rightarrow S^n$, $\tilde{H}_i(S^n - h(D^k)) = 0$ for all i .*
 (b) *For an embedding $h: S^k \rightarrow S^n$ with $k < n$, $\tilde{H}_i(S^n - h(S^k))$ is $\mathbb{Z}$ for $i = n - k - 1$ and 0 otherwise.*

As a special case of (b) we have the Jordan curve theorem: A subspace of S^2 homeomorphic to S^1 separates S^2 into two complementary components, or equivalently, path-components since open subsets of S^n are locally path-connected. One could just as well use $\mathbb{R}^2$ in place of S^2 here since deleting a point from an open set in S^2 does not affect its connectedness. More generally, (b) says that a subspace of S^n homeomorphic to S^{n-1} separates it into two components, and these components have the same homology groups as a point. Somewhat surprisingly, there are embeddings where these complementary components are not simply-connected as they are for the standard embedding. An example is the Alexander horned sphere in S^3 which we describe in detail following the proof of the proposition. These complications involving embedded S^{n-1} 's in S^n are all local in nature since it is known that any locally nicely embedded S^{n-1} in S^n is equivalent to the standard $S^{n-1} \subset S^n$, equivalent in the sense that there is a homeomorphism of S^n taking the given embedded S^{n-1} onto the standard S^{n-1} . In particular, both complementary regions are homeomorphic to open balls. See [Brown 1960] for a precise statement and proof. When $n = 2$ it is a classical theorem of Schoenflies that all embeddings $S^1 \hookrightarrow S^2$ are equivalent.

By contrast, when we come to embeddings of S^{n-2} in S^n , even locally nice embeddings need not be equivalent to the standard one. This is the subject of knot theory, including the classical case of knotted embeddings of S^1 in S^3 or $\mathbb{R}^3$. For embeddings of S^{n-2} in S^n the complement always has the same homology as S^1 , according to the theorem, but the fundamental group can be quite different. In spite of the fact that the homology of a knot complement does not detect knottedness, it is still possible to use homology to distinguish different knots by looking at the homology of covering spaces of their complements.

Proof: We prove (a) by induction on k . When $k = 0$, $S^n - h(D^0)$ is homeomorphic to $\mathbb{R}^n$, so this case is trivial. For the induction step it will be convenient to replace the domain disk D^k of h by the cube I^k . Let $A = S^n - h(I^{k-1} \times [0, 1/2])$ and let